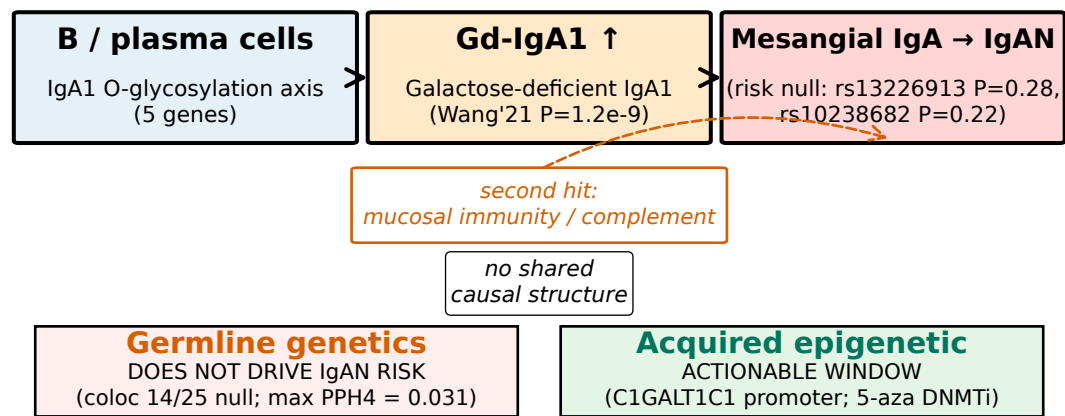
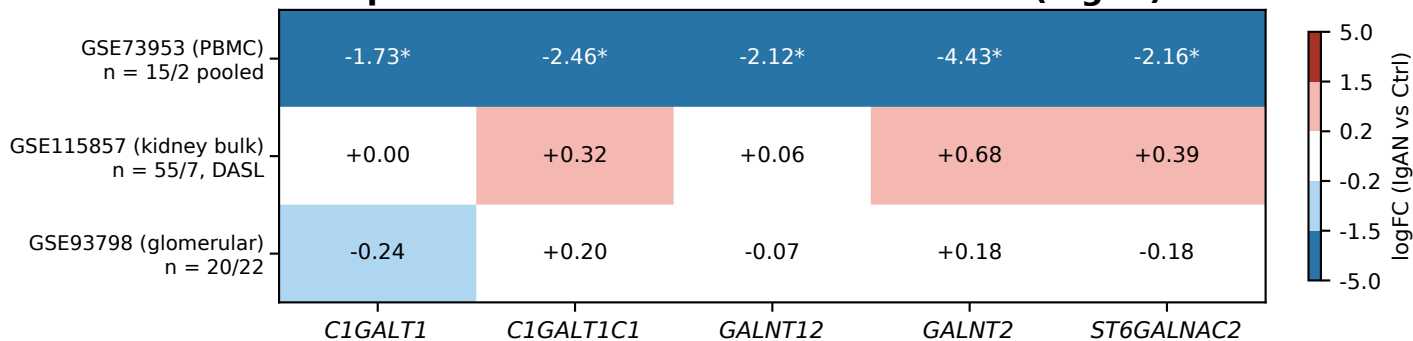


a Conceptual framework of the axis



b Expression direction across 3 datasets (logFC)



* effect-size cutoff (P<0.05 & |logFC|>1), not nominal significance; PBMC uniformly down (5/5); kidney directionally separated

c L0-L8 evidence chain (9 layer conclusions)

Layer	Theme	Key conclusion	Q	Fig
L0	Expression	5 genes down in PBMC (GSE73953; -1.7 to -4.4)	Q1	Fig1b
L1	Cell-type cis-eQTL	C1GALT1 B_naive P=9.4e-6; GTEx P=5.6e-12	Q1	Fig2
L2	Coloc with IgAN	Coloc 14/25 OK; max PPH4=0.031; alleles null	Q3	Fig3
L3	Methylation	5 gene-body CpG (no promoter); MR null	Q2	Fig4
L4	cis-eQTL lookup	10/10 axis-region top variants not IgAN-sig.	Q3	Tbl.S4
L5	EAS heterogeneity	rs13226913 EAS 0.93 vs EUR 0.42; rs7856182 inv.	Q4	Fig5
L6	Intervention	C1GALT1C1=6.0 > C1GALT1=4.5; DNMTi [22, 23]		Fig6
L7	Cell-state check	C1GALT1C1 detection 3.1→5.2% (naive→plasma)	Q1	Fig.S4
L8	Upstream regulon	Negative vs matched null; Sp/KLF-dense island	Q2	Fig.S5-6